Native Markdown-to-Word for AI-Assisted Writing

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Abstract

Academic writing in the age of AI assistants needs a text-first source of truth. This demo shows the same manuscript compiled to a publication-ready Word file: numbered headings, a three-line table, a captioned figure, a centred equation with a right-aligned number, and an automatically generated bibliography.

Markdown, Word, typesetting, citations, reproducible research

Introduction

Writing with an AI assistant means writing in Markdown: plain text, versioned in Git, diffable line by line (Smith & Doe, 2024). The deliverable, however, is still a `.docx` that a journal or an institution will accept (Doe, 2023). This demo is the shortest path between the two.

Method

The pipeline reads a fixed Markdown dialect, builds a document model, and writes OOXML directly — no external binary is called. Citations come from a `.bib` file and a CSL style; everything else comes from a plain-text requirements file.

Table 1. What each input file contributes

Input	Role	Edited by
`paper.md`	content and structure	the author
`refs.bib`	bibliography entries	reference manager
`format.txt`	fonts, spacing, page setup	the institution

Note. Only the first row is written by hand; the bibliography is generated.

Equations and cross-references

The model is deliberately simple:

$$y = \beta_0 + \sum_k \beta_k \, x_k + \varepsilon \tag{1}$$

As shown in Equation (1), the relation is linear in the coefficients.

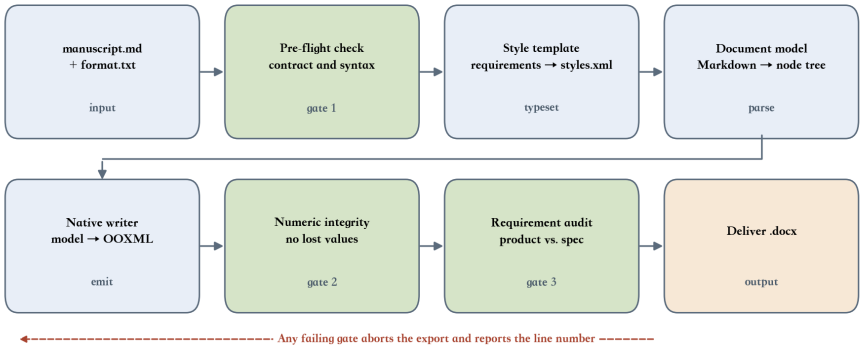


Figure 1. From Markdown source to a Word deliverable

Discussion

Two properties matter more than any single feature. First, the source stays readable: a paragraph is one line, and a style is a line prefix, so nothing has to be remembered beyond `#` and `%%`. Second, the result is deterministic: the same inputs produce the same document, which is what makes a diff meaningful (Lee, 2022).

References

Doe, J. (2023). Markdown as a source of truth. *Text Studies*, *12*, 33–48.

Lee, K. (2022). *Binary No More* (Version, Issue). Academic Press.

Smith, J., & Doe, J. (2024). Structured writing for machines. *Journal of Documentation*, *80*(2), 101–120.